Multimedia Signal processing

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September, 10 2013

1. **Question (8 pts)**:
   Consider a sequence of symbols *abbbbaaacccacbccccbcacbbbbbbbc* emitted by a source X.
   (a) Learn the p.m.f. of the source from the observed data and compute the entropy of the source (assuming a memoryless source).
   (b) To exploit source memory, group symbols in blocks of size $N = 2$ and compute $H_2(X)$.
   (c) Compute the number of bits needed to encode the sequence *bbbcaa* using Huffman coding in two cases: i) without exploiting source memory; ii) exploiting source memory.
2. **Question (5 pts)** Transform coding is a key component in most audio-visual coding standard.
   (a) Illustrate the basic principles of transform coding, providing a geometric intrepretation.
   (b) Indicate how transform coding is used in current audio-visual coding standards.
3. **Question (5 pts)**
   Unlike general audio signals, speech has some unique properties in the time-frequency domain
   (a) Illustrate the properties of speech signals.
   (b) Describe how state-of-the-art speech coders exploit such properties.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**
   (a) Let X and Y denote two discrete sources and $R(D)$ the rate-distortion function.
   - [T] [F] If X and Y are both memoryless and they have the same p.m.f., $R_X(D) = R_Y(D)$.
   - [T] [F] If the number of symbols of X is greater than the number of symbols of Y, then $R_X(0) > R_Y(0)$
   - [T] [F] $R_X(\sigma_X^2) = 0$.
   - [T] [F] Any practical coding scheme is bound to achieve an operational rate-distortion function $R_X^o(D)$ such that $R_X^o(D) \geq R_X^S(D)$, where $R_X^S(D)$ is Shannon's bound for the source X.

   (b) Consider a JPEG codec
   - [T] [F] If a JPEG compressed image I_1 is re-compressed with JPEG with the same quantization matrix to obtain I_2, then $I_1 = I_2$.
   - [T] [F] The DC coefficient is encoded by means of DPCM.
   - [T] [F] The size of the JPEG bitstream is fixed, once the spatial resolution of the image is given.

   (c) Motion estimation is a key component in any video coding architecture.
   - [T] [F] Motion estimation is typically performed in the DCT domain.
   - [T] [F] The cost of motion estimation is always quadratic in W, where W is the size of the $W \times W$ search window.
   - [T] [F] Motion estimation is performed both at the encoder and at the decoder.

5. **Question (Matlab - 10pts)**
   (a) Generate $N = 1000$ samples of a signal obtained by filtering an i.i.d. Gaussian noise signal ($\mu_z = 0$, $\sigma_z^2 = 1$) with the following filter.

$$H(z) = \frac{1}{1 - 0.9z^{-1}} \tag{1}$$

   (b) Implement a DPCM encoder, which receives as input the following parameters:
   - $\Delta \in \{0.025, 0.05, 0.1, 0.2, 0.4\}$, which is the quantization step size
   - $\alpha \in \{0.8, 0.9, 1.0, 1.1, 1.2\}$, which is a parameter used to tune the predictor $\hat{x}(n)$, i.e., $\hat{x}(n) = \alpha\tilde{x}(n-1)$, where $\tilde{x}(n)$ represents the DPCM decoded sequence.

   (c) For each value of α, plot the rate-distortion curve (Hint: approximate rate as the entropy of the quantization symbols).

   (d) Compare the rate-distortion curves and comment the result.